

AI Dress Code Compliance System

Project Description

1. Introduction

This project is an Artificial Intelligence based Dress Code Compliance System for a corporate office. The system verifies if the employees are wearing the department dress or not. It leverages Machine Learning to forecast whether an employee is compliant or non-compliant based on the information about their clothes. Flask is used for deploying the system as a web application.

2. Problem Statement

There are different policies on dress codes in corporate offices for various departments. The manual process of checking each employee is labour intensive. A trained ML model that automatically predicts whether an employee's clothing abides by the dress code rules or not is used to automate this system.

3. Dataset

The dataset was not available in Kaggle, hence a custom dataset was created using the Python script. It holds 100 records, each having 11 columns: employee_id, name, department, gender, shirt_color, shirt_type, pants_type, shoe_type, has_id_card, has_safety_gear, and is_compliant. There are 54 records that meet the criteria and 46 records that don't. There were also some null values used for pre-processing.

There is a dress code in place for the Department. The Department has dress code rules.

IT Department: Smart casual – formal shirt or polo shirt/ casual shirt is OK with dress pants/ jeans or chinos.

HR Department: Formal - Formal Shirt, Dress Pants, Formal Shoes, ID Card required.

Finance Department: Formal - Formal Shirt only, Dress Pants only, Formal Shoes or Loafers and ID Card required.

Marketing Department: Business Casual - Formal Shirt, Polo or Casual Shirt with Dress Pants or Chinos.

Factory Department: Safety Required - Safety Vest required, Cargo Pants or Dress Pants, Safety Boots required, ID Card and Safety Gear required.

Management: Executive Formal - Formal Shirt in White, Blue or Black only, Dress Pants, Formal Shoes, ID Card required.

5. Methodology

Step 1- Data Loading:

CSV file is loaded using Pandas library. Step 1- Data Loading: CSV file is loaded using Pandas library. Basic exploration is done including checking rows, columns, null values and data types.

Step 2 - Data Preprocessing:

Mode is used for filling the null values. The unnecessary columns such as employee_id and name are removed. Factorization is used to convert categorical text columns to integer.

Step 3 - Train Test Split:

The data is divided into 70% training data & 30% testing data.

Step 4 - Model Training:

Three classifiers are trained and evaluated. Random Forest gave 96.67% accuracy. Decision Tree was found to be 100% accurate. KNN gave 83.33% accuracy.

The accuracy, precision, recall and F1 scores are plotted in step 5 – Visualization: Line graph and bar graph.

Step 6 – Model Saving:

Pickle is used to save the best model for use in the Flask app.

6. Flask Web Application

Flask is used to deploy the trained model. The application consist of 3 pages.

Home Page: A form in which the user enters employee information (department, shirt type, pants type, shoe type, ID card and safety equipment status).

Result Page: Provides the employee with the results of the model prediction – either COMPLIANT or NON-COMPLIANT.

Rules Page: This will show the dress code rules for each department.

7. Technologies Used

Python - Programming language.

Loading and preprocessing data with Pandas.

NumPy - Numerical operations.

The scikit-learn package - Machine Learning classifiers.

Matplotlib - Graphs and visualization.

Save and load model. Pickle - Save and load model.

Flask - A lightweight Web application framework.

HTML and CSS - Frontend design.

8. Project Structure

The files are app.py, train_model.py and dress_code_dataset.csv in the AI Project folder. There are two .pkl files in the models folder: dress_code_model.pkl and factorize_mappings.pkl. Inside static folder you will find style.css, line_graph.png and bar_graph.png. There are three templates in the templates folder: index.html, result.html and rules.html.

9. How to Run

Step 1: Install packages with pip: pip install flask pandas scikit-learn matplotlib numpy.

Step 2: Execute the train_model.py script to train the model and create graphs.

Step 3: Run app.py to start the Flask server.

Step 4: Open browser and go to <http://127.0.0.1:5000>.

10. Future Improvements

This system can be enhanced in the future by implementing an image detection system based on the YOLO algorithm and camera in order to monitor the dress code in real time. This would mean that clothing could be detected automatically from pictures rather than having to enter the form in by hand. An employee record and compliance history can also be stored in a database.